

2019 Update to SEDAR 38 South Atlantic and Gulf of Mexico King Mackerel

South Atlantic

In addition to the research recommendations documented during SEDAR 38, additional recommendations were noted during the update assessment. Research aimed at improving the documentation of data series formatting, including index standardization, for SS3 would improve modeling efficiency. This includes statistical coding for consistent database querying and data processing. An evaluation of alternative age references, or age-specific time series, for the SEAMAP fishery independent survey was recommended by the data providers and noted by the analyst for future assessments. An analysis of the effect of excluding sublegal fish size observations on the assessment should be undertaken. Information on the age-composition of discarded fish from all fleets is needed to validate the assumption of exclusively age-0 discards. The conditional age-at-length data had a significant influence on recent recruitment estimates.

Future research assessments should evaluate model sensitivity to the age-data and explore alternative parameterizations (such as inverse age-length key), as the fleet coverage was suboptimal with zero information available for several fleets and years.

Gulf of Mexico

- The assessment should be moved to the most updated version of SS, which, among other improvements, has the means to explicitly model a bycatch-only fishery.
- The reasons for the conflicting signals in the data requires further investigation.
- The assessment should take full advantage of the upcoming shrimp bycatch estimation workshop and any results or recommendation resulting from it.
- The number of samples of lengths of discarded, undersized fish needs to be increased. This should add stability to model as well as increase the accounting of dead discards.